

# Daily Algebra Drill #5 — Answers & Investigations

Sheet by David Akinyele-Aje

| Section | Topic                    | Questions | Target time  |
|---------|--------------------------|-----------|--------------|
| A       | Rapid Recognition        | 10        | 2 min 30 sec |
| B       | Manipulation Drills      | 10        | 8 min        |
| C       | Substitution & Structure | 8         | 10 min       |
| D       | Challenge                | 5         | 15 min       |

New toolkit today: Partial fractions with quadratic denominators · Functional equations · Cauchy–Schwarz (Engel form) · Conjugate surd pairs.

## Section A Rapid Recognition

1. Simplify  $\frac{x^2 - 9}{x^2 + 5x + 6}$ .

**Answer:**  $\frac{x - 3}{x + 2}$

**Method:**  $(x - 3)(x + 3)/[(x + 2)(x + 3)]$ . Cancel  $(x + 3)$ . Undefined at  $x = -3$  and  $x = -2$ .

**Investigate further:** The cancelled factor  $(x + 3)$  creates a *removable discontinuity* (hole) at  $x = -3$ . Sketch the graph of both the original and simplified form. What is the  $y$ -value of the hole?

2. Rationalise:  $\frac{4}{\sqrt{7} - \sqrt{3}}$ .

**Answer:**  $\sqrt{7} + \sqrt{3}$

**Method:** Multiply by  $\frac{\sqrt{7} + \sqrt{3}}{\sqrt{7} + \sqrt{3}}$ : denominator =  $7 - 3 = 4$ . Cancel the 4.

**Investigate further:** Rationalising is conjugate multiplication. The pattern  $(\sqrt{a} - \sqrt{b})(\sqrt{a} + \sqrt{b}) = a - b$  applies whenever you see a two-term surd denominator. Use it to simplify  $\frac{1}{\sqrt{n+1} - \sqrt{n}}$  and telescope  $\sum_{n=1}^{99} \frac{1}{\sqrt{n+1} - \sqrt{n}}$ .

3. Given  $x + y = 3$ ,  $xy = -4$ , find  $(x - y)^2$ .

**Answer:** 25

**Method:**  $(x - y)^2 = (x + y)^2 - 4xy = 9 + 16 = 25$ .

**Investigate further:** So  $x - y = \pm 5$ . Combined with  $x + y = 3$ :  $x = 4, y = -1$  or  $x = -1, y = 4$ . Verify  $xy = -4$ . ✓ Now: without finding  $x$  and  $y$ , find  $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$ .

4. Factorise:  $x^3 - x^2 - 4x + 4$ .

**Answer:**  $(x - 1)(x - 2)(x + 2)$

**Method:** Group:  $x^2(x - 1) - 4(x - 1) = (x - 1)(x^2 - 4) = (x - 1)(x - 2)(x + 2)$ .

**Investigate further:** The grouping  $x^2(x-1) - 4(x-1)$  works because both groups share the factor  $(x-1)$ . For which cubics  $x^3 + ax^2 + bx + c$  does this grouping strategy work? (When  $c = ab$ , so that  $x^3 + ax^2 + bx + ab = x^2(x+a) + b(x+a)$ .) Verify for this example:  $a = -1, b = -4, c = 4 = (-1)(-4)$ .  
✓

5. Without expanding, find the coefficient of  $x^2$  in  $(1+x)^4$ .

**Answer:** 6

**Method:**  $\binom{4}{2} = 6$ .

**Investigate further:** Pascal's triangle row 4: 1, 4, 6, 4, 1. Memorise rows 0–6. Then: find the coefficient of  $x^3$  in  $(2+3x)^5$  using the binomial theorem:  $\binom{5}{3}(2)^2(3x)^3 = 10 \cdot 4 \cdot 27x^3 = 1080x^3$ .

6. Write the quadratic with roots  $\alpha, \beta$  where  $\alpha + \beta = -3, \alpha\beta = -10$ .

**Answer:**  $x^2 + 3x - 10 = 0$

**Method:**  $x^2 - (\alpha + \beta)x + \alpha\beta = x^2 - (-3)x + (-10) = x^2 + 3x - 10$ .

**Investigate further:** Factorise:  $(x+5)(x-2) = 0$ , roots  $-5$  and  $2$ . Check sum  $= -3$ , product  $= -10$ . ✓ Now find the quadratic with roots  $\alpha^2$  and  $\beta^2$  using only the sums and products (without finding  $\alpha, \beta$ ): sum  $= \alpha^2 + \beta^2 = 9 + 20 = 29$ , product  $= 100$ .

7. Simplify:  $\sqrt{50} - \sqrt{18} + \sqrt{8}$ .

**Answer:**  $4\sqrt{2}$

**Method:**  $5\sqrt{2} - 3\sqrt{2} + 2\sqrt{2} = 4\sqrt{2}$ .

**Investigate further:** Always reduce surds to  $k\sqrt{n}$  with  $n$  square-free before combining. The technique:  $\sqrt{50} = \sqrt{25 \cdot 2} = 5\sqrt{2}$ . Simplify  $\sqrt{75} + \sqrt{48} - \sqrt{27}$  and  $\sqrt{200} - \sqrt{162} + \sqrt{32}$ .

8. Given  $a + b = 7, a^2 + b^2 = 29$ , find  $a^3 + b^3$ .

**Answer:** 133

**Method:**  $ab = \frac{(a+b)^2 - (a^2 + b^2)}{2} = \frac{49 - 29}{2} = 10$ . Then  $a^3 + b^3 = 7(29 - 10) = 7 \times 19 = 133$ .

**Investigate further:** Note  $133 = 7 \times 19$ . Also  $a^3 + b^3 = (a+b)[(a+b)^2 - 3ab] = 7(49 - 30) = 7 \times 19$ . Verify  $a = 5, b = 2$  satisfies both conditions and  $5^3 + 2^3 = 133$ . ✓

9. Solve:  $|2x - 3| = 7$ .

**Answer:**  $x = 5$  or  $x = -2$

**Method:**  $2x - 3 = 7 \Rightarrow x = 5; 2x - 3 = -7 \Rightarrow x = -2$ . Draw the V-shape:  $y = |2x - 3|$  meets  $y = 7$  at two points.

**Investigate further:** The solution set  $\{x : |2x - 3| = 7\} = \{-2, 5\}$ . Now solve  $|2x - 3| = |x + 4|$  by splitting into cases  $2x - 3 = x + 4$  and  $2x - 3 = -(x + 4)$ .

10. Find the minimum of  $\frac{(x+4)^2}{x}$  for  $x > 0$ .

**Answer:** 16

**Method:**  $\frac{(x+4)^2}{x} = x + 8 + \frac{16}{x} \geq 2\sqrt{16} + 8 = 16$  by AM–GM on  $x + \frac{16}{x} \geq 8$ . Equality at  $x = 4$ .

**Investigate further:** Alternatively write  $\frac{(x+4)^2}{x} = (\sqrt{x} + \frac{4}{\sqrt{x}})^2$ . Then minimise  $\sqrt{x} + \frac{4}{\sqrt{x}}$  using AM–GM:  $\geq 2\sqrt{4} = 4$ . Squaring:  $\geq 16$ . This substitution approach ( $t = \sqrt{x}$ ) is faster for recognising the perfect square form.

## Section B Manipulation Drills

1. Partial fractions:  $\frac{5x+1}{(x+1)(x-2)}$ .

**Answer:**  $\frac{4/3}{x+1} + \frac{11/3}{x-2}$ , or  $\frac{4}{3(x+1)} + \frac{11}{3(x-2)}$

**Method:**  $5x+1 = A(x-2) + B(x+1)$ .  $x=2$ :  $11 = 3B \Rightarrow B = \frac{11}{3}$ .  $x=-1$ :  $-4 = -3A \Rightarrow A = \frac{4}{3}$ .

**Investigate further:** Partial fractions are the algebraic inverse of combining fractions. They are essential for integrating rational functions and for telescoping sums. Now decompose  $\frac{2x^2+3}{(x-1)(x+1)(x+2)}$  — note the degree of numerator is less than the degree of denominator, so the form is  $\frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{x+2}$ .

2. Partial fractions:  $\frac{3}{x(x^2+3)}$ .

**Answer:**  $\frac{1}{x} - \frac{x}{x^2+3}$

**Method:**  $3 = A(x^2+3) + (Bx+C)x$ .  $x=0$ :  $3 = 3A \Rightarrow A = 1$ . Coefficient of  $x^2$ :  $0 = A + B \Rightarrow B = -1$ . Coefficient of  $x$ :  $0 = C$ . Result:  $\frac{1}{x} - \frac{x}{x^2+3}$ .

**Investigate further:** The irreducible quadratic denominator  $x^2+3$  has roots  $\pm i\sqrt{3}$ , which are complex. This means we *cannot* factor further over  $\mathbb{R}$ , so the numerator over  $x^2+3$  must be linear ( $Bx+C$ ). This is the key rule: linear numerator over irreducible quadratic. Practice with  $\frac{x^2+1}{x(x^2+x+1)}$ .

3. Show  $\frac{1}{\sqrt{n} + \sqrt{n+1}} = \sqrt{n+1} - \sqrt{n}$ ; hence evaluate  $\sum_{k=0}^8 \frac{1}{\sqrt{k} + \sqrt{k+1}}$ .

**Answer:** 3

**Method:** Rationalise:  $\frac{\sqrt{n+1}-\sqrt{n}}{(\sqrt{n+1}-\sqrt{n})(\sqrt{n+1}+\sqrt{n})} = \sqrt{n+1} - \sqrt{n}$ . The sum telescopes:  $(\sqrt{1} - \sqrt{0}) + (\sqrt{2} - \sqrt{1}) + \dots + (\sqrt{9} - \sqrt{8}) = \sqrt{9} - \sqrt{0} = 3$ .

**Investigate further:** This is the classic rationalise-then-telescope technique. Now evaluate  $\sum_{k=1}^n \frac{1}{\sqrt{k} + \sqrt{k+1}}$  in closed form (answer:  $\sqrt{n+1} - 1$ ), and find the smallest  $n$  for which this sum exceeds 9.

4. Factorise  $a^4 + a^2 + 1$  over  $\mathbb{Z}$ .

**Answer:**  $(a^2 + a + 1)(a^2 - a + 1)$

**Method:** Add and subtract  $a^2$ :  $a^4 + 2a^2 + 1 - a^2 = (a^2 + 1)^2 - a^2 = (a^2 + a + 1)(a^2 - a + 1)$ .

**Investigate further:** Note  $a^4 + a^2 + 1 = \frac{a^6-1}{a^2-1}$  (geometric series), so this factorisation reflects the cyclotomic structure of  $x^6 - 1$ . Verify that the roots of  $a^2 + a + 1 = 0$  and  $a^2 - a + 1 = 0$  are the primitive 3rd and 6th roots of unity respectively.

5. Express  $(a - b)^4$  in terms of  $s = a + b$  and  $p = ab$ .

**Answer:**  $(s^2 - 4p)^2$

**Method:**  $(a - b)^2 = (a + b)^2 - 4ab = s^2 - 4p$ . Then  $(a - b)^4 = [(a - b)^2]^2 = (s^2 - 4p)^2$ .

**Investigate further:** Find  $(a - b)^6$  and  $(a - b)^8$  in terms of  $s$  and  $p$ . Note the pattern:  $(a - b)^{2k} = (s^2 - 4p)^k$ . Now express  $a^4 + b^4$  in terms of  $s$  and  $p$ :  $a^4 + b^4 = (a^2 + b^2)^2 - 2a^2b^2 = [(s^2 - 2p)^2 - 2p^2]$ .

6. Simplify  $\frac{x^2 - 2x + 1}{x^2 + 2x - 3} \div \frac{x^2 - 1}{x^2 + 4x + 3}$ .

**Answer:** 1

**Method:** Factor everything:  $\frac{(x-1)^2}{(x-1)(x+3)} \cdot \frac{(x+1)(x+3)}{(x-1)(x+1)} = \frac{x-1}{x+3} \cdot \frac{x+3}{x-1} = 1$ .

**Investigate further:** This simplification to 1 means the two rational expressions are reciprocals of each other. Verify by noting that the second fraction is the reciprocal of the first (after simplification). For what values of  $x$  is the original expression undefined?

7. Find the quadratic with roots  $r^2$  and  $s^2$  where  $r, s$  are roots of  $x^2 + px + q = 0$ .

**Answer:**  $x^2 - (p^2 - 2q)x + q^2 = 0$

**Method:**  $r^2 + s^2 = (r + s)^2 - 2rs = p^2 - 2q$ .  $(rs)^2 = q^2$ . Quadratic:  $x^2 - (p^2 - 2q)x + q^2 = 0$ .

**Investigate further:** This is the Tschirnhaus substitution idea: transforming the roots without solving for them. Now find the quadratic whose roots are  $r^3, s^3$  when  $r, s$  satisfy  $x^2 - 3x + 1 = 0$ . (Hint: use the fact that  $r^3 + s^3$  and  $r^3s^3$  can be found from the Vieta data.)

8. Solve:  $\frac{1}{x} + \frac{1}{y} = \frac{5}{6}$  and  $\frac{1}{x} - \frac{1}{y} = \frac{1}{6}$ .

**Answer:**  $x = 2, y = 3$

**Method:** Let  $u = \frac{1}{x}, v = \frac{1}{y}$ ;  $u + v = \frac{5}{6}, u - v = \frac{1}{6}$ . Adding:  $2u = 1 \Rightarrow u = \frac{1}{2}, v = \frac{1}{3}$ .

**Investigate further:** This substitution ( $u = 1/x$ ) converts a non-linear system to a linear one. The same trick works for systems involving  $\frac{1}{x^2}$  (substitute  $u = \frac{1}{x^2}$ ). Now solve:  $\frac{2}{x} + \frac{3}{y} = 7$  and  $\frac{4}{x} - \frac{1}{y} = 1$ .

9. Prove:  $\frac{1}{(2k-1)(2k+1)} = \frac{1}{2} \left( \frac{1}{2k-1} - \frac{1}{2k+1} \right)$ ; hence  $\sum_{k=1}^n \frac{1}{(2k-1)(2k+1)} = \frac{n}{2n+1}$ .

**Answer:** Proof by partial fractions, then telescoping.

**Method:** PF:  $\frac{1}{(2k-1)(2k+1)} = \frac{A}{2k-1} + \frac{B}{2k+1}$ . Solving:  $A = \frac{1}{2}, B = -\frac{1}{2}$ . Sum telescopes:  $\frac{1}{2} \left( 1 - \frac{1}{2n+1} \right) = \frac{n}{2n+1}$ .  $\square$

**Investigate further:** The denominator  $(2k-1)(2k+1)$  is the product of two consecutive odd numbers. The partial fraction structure is  $\frac{1}{2d}(\frac{1}{a} - \frac{1}{b})$  where  $b-a=d$ . This generalises:  $\frac{1}{(n+a)(n+b)} = \frac{1}{b-a}(\frac{1}{n+a} - \frac{1}{n+b})$ . Use this to find  $\sum_{k=1}^n \frac{1}{(k+1)(k+3)}$ .

10. If  $f(x) = \frac{ax+b}{cx+d}$  and  $f(f(x)) = x$ , what condition must  $a, b, c, d$  satisfy?

**Answer:**  $a+d=0$  (equivalently  $d=-a$ )

**Method:** Compute  $f(f(x)) = \frac{a \cdot f(x) + b}{c \cdot f(x) + d} = \frac{(a^2+bc)x + (a+d)b}{(a+d)cx + (bc+d^2)}$ . For this to equal  $x$ : need  $a^2+bc=bc+d^2$  (i.e.  $a^2=d^2$ ) and  $(a+d)b=0$  and  $(a+d)c=0$ . If  $a=-d$ , all conditions hold automatically.

**Investigate further:** Such a function is called an *involution* or a *Möbius involution*. Examples:  $f(x) = \frac{1}{x}$  ( $a=0, d=0$  degenerate),  $f(x) = \frac{-x+1}{x+1}$  (check  $a+d=0$ ). Find two more Möbius involutions and verify  $f(f(x)) = x$  for each.

## Section C Substitution & Structure

1. Solve:  $\left(x^2 + \frac{1}{x^2}\right) - 7\left(x - \frac{1}{x}\right) - 10 = 0$ .

**Answer:**  $x - \frac{1}{x} = 8$  (giving  $x = 4 \pm \sqrt{17}$ ) or  $x - \frac{1}{x} = -1$  (giving  $x = \frac{-1 \pm \sqrt{5}}{2}$ )

**Method:** Let  $u = x - \frac{1}{x}$ , so  $x^2 + \frac{1}{x^2} = u^2 + 2$ . Then  $(u^2 + 2) - 7u - 10 = 0 \Rightarrow u^2 - 7u - 8 = 0 \Rightarrow (u-8)(u+1) = 0$ .  $u=8$ :  $x^2 - 8x - 1 = 0 \Rightarrow x = 4 \pm \sqrt{17}$ .  $u=-1$ :  $x^2 + x - 1 = 0 \Rightarrow x = \frac{-1 \pm \sqrt{5}}{2}$ .

**Investigate further:** The lesson: when you see  $x^2 + \frac{1}{x^2}$  alongside  $x - \frac{1}{x}$ , use  $u = x - \frac{1}{x}$  (since  $u^2 = x^2 - 2 + \frac{1}{x^2}$ ). When you see  $x^2 + \frac{1}{x^2}$  alongside  $x + \frac{1}{x}$ , use  $v = x + \frac{1}{x}$  instead. Carefully distinguish which combination is present before substituting.

2. Using the Engel (Titu) form of Cauchy-Schwarz, prove  $\frac{a^2}{b} + \frac{b^2}{c} + \frac{c^2}{a} \geq 3$  when  $a+b+c=3$ .

**Answer:** Proved below.

**Method:** Titu:  $\frac{a^2}{b} + \frac{b^2}{c} + \frac{c^2}{a} \geq \frac{(a+b+c)^2}{b+c+a} = \frac{9}{3} = 3$ .  $\square$  Equality iff  $\frac{a}{b} = \frac{b}{c} = \frac{c}{a}$ , i.e.  $a=b=c=1$ .

**Investigate further:** The Engel/Titu form states  $\sum \frac{x_i^2}{y_i} \geq \frac{(\sum x_i)^2}{\sum y_i}$ . Prove it from the standard Cauchy-Schwarz  $(\sum u_i v_i)^2 \leq (\sum u_i^2)(\sum v_i^2)$  by setting  $u_i = \frac{x_i}{\sqrt{y_i}}$  and  $v_i = \sqrt{y_i}$ . This form is indispensable in olympiad inequality problems.

3. Solve the system:  $x^2 + xy = 12$ ,  $xy + y^2 = 6$ .

**Answer:**  $(x, y) = (2\sqrt{2}, \sqrt{2})$  and  $(-2\sqrt{2}, -\sqrt{2})$

**Method:** Factor both left sides:  $x(x+y) = 12$  and  $y(x+y) = 6$ . Neither  $y$  nor  $x+y$  can be zero (each would force  $0=6$  or  $0=12$ ), so divide:  $\frac{x}{y} = 2$ , giving  $x=2y$ . Substitute into  $xy + y^2 = 6$ :  $2y^2 + y^2 = 6 \Rightarrow y^2 = 2 \Rightarrow y = \pm\sqrt{2}$ , with  $x=2y$  matching the sign. Check:  $8+4=12 \checkmark$ ,  $4+2=6 \checkmark$ .

**Investigate further:** Dividing equations is a powerful technique when the ratio reveals a simple relationship. But division requires care: we must check we're not dividing by zero. When might

$xy + y^2 = 0$ ? Then  $y = 0$  or  $y = -x$ ; check these cases separately.

4. Find values of  $k$  for which  $kx^2 - 4x + k - 3 = 0$  has two distinct real roots.

**Answer:**  $-1 < k < 4$  and  $k \neq 0$

**Method:** For a quadratic ( $k \neq 0$ ): discriminant  $= 16 - 4k(k - 3) = 16 - 4k^2 + 12k > 0 \Rightarrow 4k^2 - 12k - 16 < 0 \Rightarrow k^2 - 3k - 4 < 0 \Rightarrow (k - 4)(k + 1) < 0$ . So  $-1 < k < 4$ . (Exclude  $k = 0$  since that makes it linear, not quadratic.)

**Investigate further:** The discriminant condition is the standard approach, but always check the leading coefficient separately. Now find the values of  $k$  for which  $kx^2 - 4x + k - 3 = 0$  has *equal* roots (one repeated real root). (Answer:  $k = -1$  or  $k = 4$ .)

5. Simplify  $\sqrt{3 + 2\sqrt{2}} + \sqrt{3 - 2\sqrt{2}}$ .

**Answer:**  $2\sqrt{2}$

**Method:**  $3 + 2\sqrt{2} = (\sqrt{2} + 1)^2$ , so  $\sqrt{3 + 2\sqrt{2}} = \sqrt{2} + 1$ . Similarly  $3 - 2\sqrt{2} = (\sqrt{2} - 1)^2 \Rightarrow \sqrt{3 - 2\sqrt{2}} = \sqrt{2} - 1$ . Sum  $= 2\sqrt{2}$ .

**Investigate further:** The denesting technique: to simplify  $\sqrt{a + b\sqrt{c}}$ , try  $\sqrt{p} + \sqrt{q}$  where  $p + q = a$  and  $4pq = b^2c$ . For  $\sqrt{3 - 2\sqrt{2}}$ :  $p + q = 3$ ,  $4pq = 8 \Rightarrow pq = 2$ . So  $p, q$  are roots of  $t^2 - 3t + 2 = (t - 1)(t - 2) = 0$ :  $p = 2, q = 1$ . Hence  $\sqrt{2} - 1$  (taking positive root). Apply this to  $\sqrt{7 + 4\sqrt{3}}$ .

6. If  $x - \frac{1}{x} = 4$ , find  $\sqrt{x} - \frac{1}{\sqrt{x}}$  exactly.

**Answer:**  $\sqrt{x} - \frac{1}{\sqrt{x}} = \sqrt{2\sqrt{5} - 2}$ .

**Method:** Square the given relation:  $(x - \frac{1}{x})^2 = 16 \Rightarrow x + \frac{1}{x} = 2\sqrt{5}$  (positive since  $x > 0$ ). Let  $t = \sqrt{x} - \frac{1}{\sqrt{x}}$ . Then  $t^2 = x + \frac{1}{x} - 2 = 2\sqrt{5} - 2$ , so  $t = \sqrt{2\sqrt{5} - 2}$  (positive branch).

**Investigate further:** The substitution  $t = \sqrt{x} - 1/\sqrt{x}$  is a “half-power” reduction. Note the pattern:  $x - 1/x = t(\underbrace{\sqrt{x} + 1/\sqrt{x}}_{\sqrt{t^2+4}})$ . This generalises: to go from  $x \pm 1/x$  to  $x^{1/2} \pm 1/x^{1/2}$ , use this factorisation chain. Apply it to find  $\sqrt[4]{x} - 1/\sqrt[4]{x}$  when  $x - 1/x = \sqrt{5}$ .

7. Given  $a, b > 0$  with  $\frac{1}{a} + \frac{1}{b} = 1$ , show  $(a - 1)(b - 1) = 1$  and find  $\min(a + b)$ .

**Answer:**  $(a - 1)(b - 1) = 1$  always; minimum of  $a + b$  is 4, at  $a = b = 2$ .

**Method:** From  $\frac{1}{a} + \frac{1}{b} = 1$ :  $a + b = ab$ .  $(a - 1)(b - 1) = ab - a - b + 1 = (a + b) - (a + b) + 1 = 1$ .  $\square$  For minimum of  $a + b = ab$ : AM-GM gives  $ab \leq \left(\frac{a+b}{2}\right)^2$ , so  $a + b \leq \frac{(a+b)^2}{4}$ , giving  $a + b \geq 4$ , with equality at  $a = b = 2$ .

**Investigate further:** The constraint  $\frac{1}{a} + \frac{1}{b} = 1$  defines a hyperbola in the  $(a, b)$ -plane restricted to  $a, b > 0$ . The minimum of  $a + b$  subject to this constraint (answer: 4 at  $a = b = 2$ ) can also be found by substitution  $b = \frac{a}{a-1}$  and then minimising  $a + \frac{a}{a-1}$ . Try this approach.

8. Solve:  $(x^2 - x - 1)^2 = (2x + 1)^2$ .

**Answer:**  $x = 0, x = -1, x = \frac{3 \pm \sqrt{17}}{2}$

**Method:**  $A^2 = B^2 \Leftrightarrow (A - B)(A + B) = 0$ .  $A - B = x^2 - 3x - 2$  and  $A + B = x^2 + x = x(x + 1)$ .  
 $x(x + 1) = 0$ :  $x = 0$  or  $x = -1$ .  $x^2 - 3x - 2 = 0$ :  $x = \frac{3 \pm \sqrt{17}}{2}$ . Four real solutions in total.

**Investigate further:** The key move:  $A^2 = B^2 \Leftrightarrow (A - B)(A + B) = 0$ . Never expand both squares and combine — always factor as a difference of two squares first. Use this to solve  $|(x^2 - 2)^2 - (x - 1)^2| = 0$  and  $(x^2 + x)^2 = (x - 1)^2$ .

## Section D Challenge

(SMC / BMO1 difficulty)

1. Find  $f(x)$  given  $f(x) + f\left(\frac{1}{1-x}\right) = x$ .

**Answer:**  $f(x) = \frac{x^3 - x + 1}{2x(x - 1)}$

**Method:** Let the substitution  $\phi(x) = \frac{1}{1-x}$ . Note  $\phi(\phi(x)) = \phi\left(\frac{1}{1-x}\right) = \frac{1}{1-\frac{1}{1-x}} = \frac{1-x}{-x} = \frac{x-1}{x}$  and  $\phi^3(x) = x$  (three-cycle). So the three equations are:  $f(x) + f\left(\frac{1}{1-x}\right) = x$  (i)  $f\left(\frac{1}{1-x}\right) + f\left(\frac{x-1}{x}\right) = \frac{1}{1-x}$  (ii)  $f\left(\frac{x-1}{x}\right) + f(x) = \frac{x-1}{x}$  (iii) Add all three and divide by 2:  $f(x) + f\left(\frac{1}{1-x}\right) + f\left(\frac{x-1}{x}\right) = \frac{1}{2}\left(x + \frac{1}{1-x} + \frac{x-1}{x}\right)$ . Then subtract (ii):  $f(x) = \frac{1}{2}\left(x + \frac{1}{1-x} + \frac{x-1}{x}\right) - \frac{1}{1-x} = \frac{1}{2}\left(x - \frac{1}{1-x} + \frac{x-1}{x}\right) = \frac{x^3 - x + 1}{2x(x-1)}$ . Check at  $x = 2$ :  $f(2) + f(-1) = \frac{7}{4} + \frac{1}{4} = 2$ . ✓

**Investigate further:** The three-cycle  $x \rightarrow \frac{1}{1-x} \rightarrow \frac{x-1}{x} \rightarrow x$  is a well-known Möbius transformation cycle. Any functional equation on such a cycle can be solved by the add-then-subtract method. Find all  $f$  satisfying  $f(x) + 2f(1-x) = 3x$ .

2. Prove the Cauchy–Schwarz inequality in Engel form; find  $\min\left(\frac{1}{x} + \frac{4}{1-x}\right)$  for  $0 < x < 1$ .

**Answer:** Minimum is 9 at  $x = \frac{1}{3}$ .

**Method:** *Proof:* By standard C-S:  $\left(\sum \frac{a_i^2}{b_i}\right)(\sum b_i) \geq (\sum a_i)^2$ , giving  $\sum \frac{a_i^2}{b_i} \geq \frac{(\sum a_i)^2}{\sum b_i}$ . Apply with  $a_1 = 1, b_1 = x, a_2 = 2, b_2 = 1 - x$ :  $\frac{1}{x} + \frac{4}{1-x} \geq \frac{(1+2)^2}{x+(1-x)} = 9$ . Equality when  $\frac{1/1}{x/(1)} = \frac{2/(1)}{(1-x)/1}$ , i.e.  $\frac{1}{x} = \frac{2}{1-x} \Rightarrow x = \frac{1}{3}$ .

**Investigate further:** The Engel form inequality  $\frac{a^2}{x} + \frac{b^2}{y} \geq \frac{(a+b)^2}{x+y}$  is equivalent to  $(ay - bx)^2 \geq 0$ . Prove this algebraically. Then use the Engel form to prove that for positive  $a, b, c$ :  $\frac{a^2}{b+c} + \frac{b^2}{c+a} + \frac{c^2}{a+b} \geq \frac{a+b+c}{2}$  (Nesbitt variant).

3. Find all  $f : \mathbb{R} \rightarrow \mathbb{R}$  satisfying  $f(x + y) = f(x) + f(y)$  and  $f(1) = 3$ .

**Answer:**  $f(x) = 3x$  for  $x \in \mathbb{Q}$ ; with continuity assumption,  $f(x) = 3x$  for all  $x \in \mathbb{R}$ .

**Method:**  $f(n) = 3n$  for  $n \in \mathbb{Z}$  (by induction).  $f(p/q) = 3(p/q)$  for  $p/q \in \mathbb{Q}$  (from  $q \cdot f(p/q) = f(p) = 3p$ ). Without continuity or monotonicity, pathological solutions exist on  $\mathbb{R}$  (via Hamel basis). With any regularity assumption:  $f(x) = 3x$ .

**Investigate further:** Cauchy's functional equation  $f(x+y) = f(x) + f(y)$  (Cauchy 1821) is the first example where “algebraic” solutions and “analytic” solutions differ. Explore: what happens if we additionally require  $f$  to be monotone? Or bounded on an interval? Both conditions force  $f(x) = cx$ .

4. Show that among any 5 integers there exist 2 whose difference is divisible by 4. Hence prove that for any 5 integers, two of them satisfy  $a^2 \equiv b^2 \pmod{4}$ .

**Answer:** Proved below.

**Method:** Every integer has residue 0, 1, 2, or 3 (mod 4) (four classes). By the Pigeonhole Principle, among 5 integers, at least two share the same residue. Their difference  $\equiv 0 \pmod{4}$ .  $\square$  For the second part:  $a \equiv b \pmod{4} \Rightarrow a^2 - b^2 = (a-b)(a+b)$ . Since  $4 \mid (a-b)$ , we have  $4 \mid (a^2 - b^2)$ , i.e.  $a^2 \equiv b^2 \pmod{4}$ .

**Investigate further: Pigeonhole Principle:** if  $n$  objects are placed into  $k < n$  boxes, at least one box contains at least  $\lceil n/k \rceil$  objects. This is one of the most powerful tools in combinatorics and number theory. Use it to prove: among any 13 integers, two have the same remainder when divided by 12.

5. Find all integer solutions to  $x^2 - y^2 = 2024$ .

**Answer:**  $(x, y) = (\pm 507, \pm 505), (\pm 255, \pm 251), (\pm 57, \pm 35), (\pm 45, \pm 1)$  — all sign combinations, 16 solutions in total.

**Method:**  $(x-y)(x+y) = 2024 = 2^3 \times 11 \times 23$ . Write  $x-y = a$ ,  $x+y = b$  where  $ab = 2024$  and  $a, b$  have the same parity (both even, since 2024 is even). Set  $a = 2s$ ,  $b = 2t$ :  $4st = 2024 \Rightarrow st = 506 = 2 \times 11 \times 23$ . Factor pairs of 506 (both positive,  $s \leq t$ ): (1, 506), (2, 253), (11, 46), (22, 23). Each gives  $x = s + t$ ,  $y = t - s$ . Include negative values of  $x, y$ .

**Investigate further:**  $x^2 - y^2 = N$  has integer solutions iff  $N$  can be written as a product of two integers of the same parity. For odd  $N$ : every factorisation  $N = ab$  works (since odd  $\times$  odd = odd). For  $N \equiv 2 \pmod{4}$ : no solutions (since  $x^2 - y^2 = (x-y)(x+y)$  and  $x \pm y$  share parity, so the product is  $\equiv 0$  or  $3 \pmod{4}$ , never 2). For  $N \equiv 0 \pmod{4}$ : use even factorisations. Classify all  $N$  for which  $x^2 - y^2 = N$  has no solutions.

## Top 5 Patterns — Worksheet #5

- Partial fractions with irreducible quadratic denominator.** Use  $\frac{Bx+C}{x^2+px+q}$  (linear numerator) when the denominator has negative discriminant.
- Telescoping via rationalisation.**  $\frac{1}{\sqrt{n}+\sqrt{n+1}} = \sqrt{n+1} - \sqrt{n}$ . Memorise this — sums telescope to  $\sqrt{N} - \sqrt{0}$ .
- Engel/Titu form of Cauchy–Schwarz.**  $\sum \frac{a_i^2}{b_i} \geq \frac{(\sum a_i)^2}{\sum b_i}$ . The fastest route to many optimisation inequalities.
- Möbius involutions:**  $d = -a$ . If  $f(x) = \frac{ax+b}{cx+d}$  and  $d = -a$ , then  $f(f(x)) = x$ . Recognise this structure immediately.
- $A^2 = B^2 \Rightarrow (A-B)(A+B) = 0$ . Never expand both sides of a squared equation — always rewrite as a difference of two squares first.

## Common Traps — Worksheet #5

1. **Using a linear numerator for reducible quadratic denominators.**  $x^2 - 1 = (x - 1)(x + 1)$  is reducible — use  $\frac{A}{x-1} + \frac{B}{x+1}$ , not  $\frac{Bx+C}{x^2-1}$ .
2. **Forgetting the parity condition in  $x^2 - y^2 = N$ .** When  $N \equiv 2 \pmod{4}$ , there are no integer solutions. Check this first.
3. **Mixing  $x + 1/x$  and  $x - 1/x$  substitutions.** The identity  $x^2 + 1/x^2 = (x \pm 1/x)^2 \mp 2$  — the sign of the middle term depends on which substitution you choose.
4. **Assuming  $f(x) = cx$  is the only Cauchy solution.** Over  $\mathbb{R}$  without continuity, pathological solutions exist. Always state your assumptions.
5. **Dividing equations without checking for zero denominators.** In C3, dividing gives  $x/y = 2$ , but only if  $xy + y^2 \neq 0$ . Check the zero case separately.

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*“A formula you have derived yourself is worth a hundred you have memorised.”*

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Found a mistake or want to contribute a sheet?

Visit the open-source project at [github.com/The-CerealDev/speed-maths](https://github.com/The-CerealDev/speed-maths)